macOS Compliance & Conditional Access for Zero-Trust Access

Implementation Solution Guide

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Executive Summary — macOS Compliance & Conditional Access for Zero-Trust Access (Microsoft Intune)

White-label · Intune Solutions

A Mac shouldn't reach corporate resources just because someone has the password. Under **Zero Trust**, access is earned by a device that proves it is healthy — encrypted, patched, firewalled, and free of active threats. This solution builds the Mac-specific version of that gate: **macOS device compliance signals in Microsoft Intune** wired to a named **Conditional Access policy in Microsoft Entra**, so only trusted, compliant Macs get in.

This is a complete, **deploy-it-today** package written for an administrator with no prior macOS compliance or Conditional Access experience — including the safety practices (Report-only piloting, break-glass exclusions, grace periods) that let you enforce without locking anyone out.

What you get

- A plain-language **Solution Overview** of how macOS compliance signals feed Conditional Access under

Zero Trust.

- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click steps in the required order — build the named

macOS compliance policy, map it to a Report-only CA policy, validate, then expand.

- Five **PowerShell scripts** for Mac/compliance discovery, pre-enforcement validation, an

access-decision report, safe remediation, and rollout-group targeting.

- **Testing** and **Troubleshooting** guides for a confident, lockout-free rollout.

What this delivers

A named macOS compliance policy — **FileVault encryption**, **minimum macOS version**, **firewall on**, and **Microsoft Defender machine-risk** — mapped to a named Conditional Access policy that **requires a compliant device** for Mac sign-ins, piloted in **Report-only** on a small group before enforcement, with break-glass accounts excluded.

> *Mac-focused by design. For the general Conditional Access baseline (MFA, block legacy auth, break-glass) across Windows and mobile, deploy the companion **Conditional Access Integration** solution alongside this one.*

The standard deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Only compliant, trusted Macs reach corporate resources — compliance signals wired to Conditional Access under Zero-Trust — proven on a pilot before you scale.

1. Solution Overview — macOS Compliance & Conditional Access for Zero-Trust Access

The problem this solves

A Mac shouldn't reach corporate resources just because someone typed the right password. Under **Zero Trust**, every access request is verified against **who** the user is *and* **what state the device is in** — encrypted, patched, firewalled, and free of active threats. This solution builds the Mac-specific half of that gate: **macOS device compliance signals in Microsoft Intune** wired to a **named Conditional Access policy in Microsoft Entra**, so only trusted, compliant Macs get in.

> *This solution is **Mac-focused**. If you also run Windows and want the general Conditional Access baseline (require MFA, block legacy auth, break-glass accounts), deploy the separate **Conditional Access Integration** solution alongside it — this guide references those safety practices but does not re-teach them.*

The two halves and how they connect

1. **macOS device compliance (Intune).** A **compliance policy** evaluates each enrolled Mac against

rules — disk encryption (FileVault), minimum macOS version, firewall on, and Microsoft Defender **machine-risk** below a chosen threshold — and marks the Mac **Compliant** or **Not compliant**.

1. **Conditional Access (Entra).** A **CA policy** with the grant control **Require device to be

marked as compliant** consumes that signal. A compliant Mac is allowed; a non-compliant Mac is blocked (or challenged) when it tries to reach the targeted resources.

The Mac's compliance state is the bridge: Intune produces it, Entra enforces on it.

The macOS compliance signals this delivers

Signal	What it proves	Intune compliance setting
FileVault / disk encryption	Data at rest is protected if the Mac is lost or stolen	*Require FileVault* / *Encryption of data storage*
Minimum macOS version	The Mac is patched to a supported, non-vulnerable OS	*Minimum OS version*
Firewall enabled	The device rejects unsolicited inbound connections	*Firewall* (via macOS settings/compliance)
Microsoft Defender machine risk	No active/high-severity threat on the device	*Require the device to be at or under machine-risk score* (Defender for Endpoint connector)

Each signal that a Mac fails flips it **Not compliant**, and the CA policy then denies access — Zero Trust in action.

How it works, end to end

1. A Mac is enrolled in Intune and receives the macOS **compliance policy**.
2. Intune evaluates the Mac against the four signals and reports **Compliant** / **Not compliant**.
3. A user on that Mac signs in to a protected resource (e.g. Outlook on the web).
4. Entra evaluates the **Conditional Access policy** — *Require device to be marked as compliant*.
5. **Compliant** → **allowed. Not compliant** → **blocked** (Zero Trust: no healthy device, no access).
6. During pilot, the CA policy runs in **Report-only** so you can see *what would happen* before it actually blocks anyone.

Named objects this solution creates (and reuses)

You will build these exact objects and reuse them by name throughout the guide:

- **Compliance policy:** `macOS - Zero-Trust Compliance`
- **Conditional Access policy:** `CA - macOS - Require Compliant Device`
- **Pilot group (Entra security group):** `macOS-ZeroTrust-Pilot`
- **(Optional) exclusion group:** `macOS-ZeroTrust-Exclusions` (for break-glass / service exceptions)

Benefits

- **Device-state enforcement for Macs** — access is gated on encryption, patch level, firewall, and threat state, not just a password.
- **Zero Trust, least privilege** — a Mac earns access only while it stays healthy.
- **Safe rollout** — Report-only mode plus a small pilot group means you validate before you block.
- **Auditable** — named policies and the included reports show exactly which Macs are compliant and which sign-ins were allowed or denied.

Limitations (be honest with buyers)

- Conditional Access requires **Microsoft Entra ID P1**.
- The **Require compliant device** control depends on **this macOS compliance policy** being deployed and Macs being **Intune-enrolled and Entra-registered** first.
- The **Defender machine-risk** signal requires **Microsoft Defender for Endpoint on macOS** onboarded and the **Intune ↔ Defender connector** turned on (a companion prerequisite, flagged below).

• Available macOS compliance settings evolve; exact setting labels can differ slightly by Intune service version. Where a label may vary, this guide flags it — verify against the live console.

• A misconfigured CA policy can block legitimate users. That is why this solution pilots in **Report-only** on a small group and keeps break-glass/exclusion accounts out of scope.

The standard deployment model

> ***Assess*** → ***Configure*** → ***Pilot*** → ***Validate*** → ***Expand.*** *Build the compliance policy → map it to a Report-only CA policy on the pilot group → confirm compliant Macs are allowed and non-compliant Macs are (would-be) blocked → enable enforcement → expand to all Macs.*

2. Requirements Guide — macOS Compliance & Conditional Access for Zero-Trust Access

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites (Entra tenant, Intune subscription, admin roles, network access, a pilot group) are in `global-requirements-intune-solutions.md` — confirm those first.

> *Note: the category global requirements reference a *Windows* baseline (G6). For this solution that requirement is replaced by the **supported macOS baseline** in 2.4 below.*

2.1 Microsoft Entra ID P1 licensing

Description: Microsoft Entra ID P1 (included with Microsoft 365 E3/E5 and Business Premium). **Why:** Conditional Access is a P1 feature; without it you cannot create the CA policy that enforces the compliance signal. **Verify:** Microsoft 365 admin center → **Billing** → **Licenses** — a plan that includes Entra ID P1 is assigned to the pilot users. **Implement:** Assign Microsoft 365 E3/E5 or Business Premium to the targeted users.

2.2 Macs enrolled in Intune and Entra-registered

Description: The target Macs are enrolled in Microsoft Intune and appear as devices in Entra. **Why:** A compliance policy only reaches enrolled Macs, and the **Require compliant device** control only works if the Mac is registered in Entra so its compliance state can be read at sign-in. **Verify:** Intune → **Devices** → **All devices** — the Mac appears with OS **macOS**; run `MACCA-Discovery.ps1` to list enrolled Macs and their current compliance state. **Implement:** Enroll Macs via **Automated Device Enrollment (ADE)** or the Company Portal / user enrollment, so they become Intune-managed and Entra-registered.

2.3 Microsoft Defender for Endpoint on macOS (for the machine-risk signal)

Description: Defender for Endpoint deployed and onboarded on the Macs, with the **Intune ↔ Defender for Endpoint connector** enabled. **Why:** The **machine-risk** compliance signal reads each Mac's Defender risk score; without the connector, that signal cannot be evaluated and should be omitted. **Verify:** Intune → **Endpoint security** → **Microsoft Defender for Endpoint** — the macOS connection status is **Enabled** and *Connect macOS devices … to Microsoft Defender for Endpoint* is **On**. **Implement:** Deploy Defender for

Endpoint on macOS (companion **Defender for Endpoint (macOS) onboarding** solution), then turn on the connector. *Flag: if you are not using Defender on Macs yet, build the policy with the first three signals and add the machine-risk rule later.*

2.4 Supported macOS baseline

Description: Macs on a currently supported macOS version (a recent major release with security updates). **Why:** The **Minimum OS version** compliance rule assumes the fleet can meet a supported floor; set the floor to a version your fleet can actually reach so you do not block healthy Macs. **Verify:** On a Mac, → **About This Mac**; or Intune → device → **Hardware** → **Operating system version**. Run `MACCA-Discovery.ps1` to see the OS-version spread across enrolled Macs. **Implement:** Update Macs to a supported release (e.g. via an Intune macOS **update policy** or the built-in Software Update) before setting a high minimum-version floor.

2.5 FileVault posture (encryption signal)

Description: Macs can have FileVault enabled (either already on, or enforced via an Intune disk-encryption policy). **Why:** The encryption compliance rule marks a Mac non-compliant if FileVault is off; if you have no way to turn it on, users will be blocked with no path to remediate. **Verify:** Intune → **Endpoint security** → **Disk encryption** — a FileVault policy exists/targets the Macs; or `MACCA-Discovery.ps1` reports encryption state. **Implement:** Create an Intune **Disk encryption (FileVault)** policy so Macs can become — and stay — encrypted, then apply the compliance rule.

2.6 Pilot Entra security group

Description: A Microsoft Entra security group for the pilot Macs/users, named `macOS-ZeroTrust-Pilot`. **Why:** Both the compliance policy and the CA policy are targeted at the pilot first, never the whole fleet. **Verify:** Entra → **Groups** — `macOS-ZeroTrust-Pilot` exists; or use `MACCA-Set-ScopeTargeting.ps1` (default `-WhatIf`) to preview membership. **Implement:** Entra → **Groups** → **New group** → **Security** → `macOS-ZeroTrust-Pilot` → add pilot members. `MACCA-Set-ScopeTargeting.ps1` can build this membership from your Mac inventory.

2.7 Sign-in log / reporting access

Description: The operator can read Entra sign-in logs and Conditional Access results. **Why:** Report-only impact and allow/block decisions are reviewed in the sign-in logs before and after enforcement. **Verify:** Entra → **Monitoring** → **Sign-in logs** loads and shows a **Report-only** / **Conditional Access** tab on a sign-in. **Implement:** Grant the operator **Security Reader** or **Reports Reader** in Entra.

2.8 Break-glass / exclusion accounts (lockout safety)

Description: Emergency-access accounts (and any needed service exceptions) that are **excluded** from the CA policy. **Why:** A device-based CA policy *can* block people; break-glass accounts guarantee you can always get back in. **Verify:** Entra → **Users** — dedicated emergency accounts exist and are excluded from CA policies. **Implement:** Follow the **Conditional Access Integration** solution's break-glass guidance (two cloud-only Global Admins with offline passwords), and exclude them from `CA - macOS - Require Compliant Device`.

Checklist

- Entra ID P1 licensing assigned to pilot users
- Target Macs enrolled in Intune and Entra-registered — `MACCA-Discovery.ps1`
- Defender for Endpoint (macOS) + connector enabled *(or plan to add the machine-risk rule later)*
- Supported macOS baseline achievable across the pilot Macs
- FileVault can be enabled (disk-encryption policy in place)
- `macOS-ZeroTrust-Pilot` security group created
- Sign-in log / Report-only access for the operator
- Break-glass / exclusion accounts identified and excluded from the CA policy

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — macOS Compliance & Conditional Access for Zero-Trust Access

Follow these steps **in order**. You work in two consoles:

- **Intune admin center** — `https://intune.microsoft.com` (the macOS compliance policy).
- **Microsoft Entra admin center** — `https://entra.microsoft.com` → **Protection → Conditional Access** (the CA policy).

> ***Model:*** *Build compliance → create the CA policy in* ***Report-only*** *on the pilot → review → enable → expand.* ***Never*** *point a new device-based CA policy straight at all users.*

Objects you will create and reuse by name: `macOS - Zero-Trust Compliance` (compliance policy) · `CA - macOS - Require Compliant Device` (CA policy) · `macOS-ZeroTrust-Pilot` (pilot group) · `macOS-ZeroTrust-Exclusions` (optional).

Step 1 — Confirm the pilot group exists

1. Entra → **Groups**. Confirm `macOS-ZeroTrust-Pilot` exists (from Requirements 2.6). If not, **New group** → **Security** → name it `macOS-ZeroTrust-Pilot` → add the pilot Macs/users → **Create**.
1. *(Optional)* Create `macOS-ZeroTrust-Exclusions` the same way for any service/exception accounts you will exclude from the CA policy alongside your break-glass accounts.

Step 2 — (Optional) Confirm the Defender connector for the machine-risk signal

1. Intune → **Endpoint security** → **Microsoft Defender for Endpoint**.

2. Confirm **Connection status: Enabled** and **Connect macOS devices … to Microsoft Defender for Endpoint **is** On**.

1. If Defender on macOS is not in place yet, **skip the machine-risk rule** in Step 3.6 and add it later — the other three signals still deliver a strong Zero-Trust gate.

Step 3 — Create the macOS compliance policy `macOS - Zero-Trust Compliance`

1. Intune → **Devices** → **Compliance** (Compliance policies) → **+ Create policy**.

2. **Platform: macOS**. Click **Create**.

3. **Name:** `macOS - Zero-Trust Compliance`. **Description:** *Zero-Trust device health for Macs — encryption, OS, firewall, Defender risk.* → **Next**.

1. **Compliance settings** → **Device Health / System Security:**

• **Firewall** → **Require** (or **Enable** — the label may read *Firewall* under System Security).

*Flag: exact grouping of the firewall setting can vary by Intune version; if you don't see it under Device Health, look under **System Security**.*

1. **Encryption:**

• **Require encryption of data storage on device** → **Require** (this is FileVault on macOS).

1. **System Security** → **Operating system version:**

• **Minimum OS version** → set a floor your fleet can meet (e.g. a current supported major release). *Flag: choose a version verified reachable via `MACCA-Discovery.ps1` — too high a floor blocks healthy Macs.*

1. **Microsoft Defender for Endpoint (machine risk)** — *only if Step 2 connector is On:*

• **Require the device to be at or under the machine-risk score** → choose **Medium** (blocks High risk) to start; you can tighten to **Low** later.

1. **Actions for noncompliance** → keep the default **Mark device noncompliant** but set it to a **grace period** (e.g. *Immediately* for pilot, then a few days for production — see Best Practices) so users get a warning window before being blocked. Optionally add **Send email to end user**. → **Next**.

1. **Scope tags** → default → **Next**.

2. **Assignments** → **Included groups** → **Add groups** → select `macOS-ZeroTrust-Pilot`. → **Next**.

3. **Review + create** → **Create**. The policy `macOS - Zero-Trust Compliance` now evaluates the pilot Macs.

Step 4 — Let compliance evaluate, then confirm signals

1. Give the pilot Macs time to sync (Company Portal → **check status**, or wait for the sync cycle).
2. Intune → **Devices** → **Compliance** → `macOS - Zero-Trust Compliance` → **Device status** — confirm pilot

Macs report **Compliant** (or see exactly which signal fails).

1. Run `MACCA-Validation.ps1` to confirm the pilot Macs are **Compliant** and Entra-registered —

the two facts the CA policy depends on.

Step 5 — Create the Conditional Access policy in Report-only

1. Entra → **Protection** → **Conditional Access** → **+ Create new policy**.
2. **Name:** `CA - macOS - Require Compliant Device`.
3. **Users: Include** → **Select users and groups** → `macOS-ZeroTrust-Pilot`.

Exclude → your **break-glass** accounts (and `macOS-ZeroTrust-Exclusions` if used). *Never omit the break-glass exclusion.*

1. **Target resources: Cloud apps** → select **Office 365** (start here; broaden later).
2. **Conditions** → **Device platforms** → **Configure = Yes** → **Include** → **Select device platforms** → **macOS.**

This scopes the policy to Macs so it never affects Windows/mobile sign-ins.

1. **Grant: Grant access** → **Require device to be marked as compliant.** Leave **Require all the

selected controls **selected.** Select**.

1. **Enable policy: Report-only**. Click **Create**.

Step 6 — Review Report-only impact

1. Have pilot users sign in normally from their Macs for a day or two.
2. Entra → **Monitoring** → **Sign-in logs** → open a pilot sign-in → **Report-only** tab — confirm

`CA - macOS - Require Compliant Device` shows a result (*would apply / satisfied / would block*).

1. Run `MACCA-Report.ps1` to summarize, per Mac, the **compliance state** and the **would-be

access decision**. Resolve any unexpected would-blocks (usually a failing signal — see Troubleshooting).

Step 7 — Prove a non-compliant Mac would be blocked

1. On a **pilot test Mac**, deliberately fail one signal (e.g. turn FileVault off on a spare test

device, or let it drop below the OS floor). Wait for Intune to mark it **Not compliant**.

1. Sign in from that Mac and check the **Report-only** tab — the policy should read **would block**.
2. Confirm with `MACCA-Report.ps1` that the non-compliant Mac maps to a **deny** decision.

(Re-enable the signal afterward so the test Mac returns to Compliant.)

Step 8 — Enable enforcement

1. Re-run `MACCA-Validation.ps1` and confirm **break-glass accounts are excluded** from `CA - macOS - Require Compliant Device`.

1. Edit the CA policy → **Enable policy** → **On** → **Save**.

2. Verify pilot users on compliant Macs still work, and that the earlier non-compliant test Mac is now actually **blocked** (not just "would block").

Step 9 — Expand to all Macs

1. Broaden the **compliance policy** assignment (Step 3.10) from `macOS-ZeroTrust-Pilot` to your all-Macs group, in waves. Use `MACCA-Set-ScopeTargeting.ps1` (default `-WhatIf`) to preview the membership change before applying it.

1. Broaden the **CA policy Users** (Step 5.3) to the all-Macs group, keeping break-glass/exclusion accounts excluded. Optionally widen **Target resources** from Office 365 to **All cloud apps**.

1. Monitor each wave with `MACCA-Report.ps1`; keep the grace period generous until the fleet settles (Best Practices).

Order of operations

```
Pilot group → (Defender connector) → Compliance policy → evaluate → CA policy (Report-only) → review → prove non-compliant blocked → enable On → expand in waves
```

5. Testing Guide — macOS Compliance & Conditional Access for Zero-Trust Access

Run these during Report-only (Implementation Step 6) and again after enabling enforcement (Step 8). Core proof: **compliant Mac → allowed; non-compliant Mac → denied.**

Test 1 — Compliance policy evaluates each signal

Do: Intune → **Devices** → **Compliance** → `macOS - Zero-Trust Compliance` → **Device status**. Run `MACCA-Validation.ps1`. **Expected:** Pilot Macs show **Compliant**, or the exact failing signal (encryption / OS / firewall / machine-risk) is named.

Test 2 — CA policy evaluates in Report-only

Do: Entra → **Monitoring** → **Sign-in logs** → open a pilot Mac sign-in → **Report-only** tab. **Expected:** `CA - macOS - Require Compliant Device` shows a result — proving the policy evaluates macOS sign-ins.

Test 3 — Compliant Mac is allowed

Do: From a **Compliant** pilot Mac, sign in to `https://outlook.office.com`. Run `MACCA-Report.ps1`. **Expected:** Access granted; the report maps this Mac to **Compliant** → **allow**.

Test 4 — Non-compliant Mac is denied

Do: On a test Mac, fail one signal (FileVault off, below the OS floor, firewall off, or elevated Defender risk). After Intune marks it **Not compliant**, attempt the same sign-in. **Expected:** Report-only logs **would block**; after enforcement the sign-in is **blocked**. `MACCA-Report.ps1` maps this Mac to **Not compliant** → **deny**.

Test 5 — Each signal independently trips compliance

Do: One at a time on a test Mac: disable FileVault; drop below the OS floor; disable the firewall; (if Defender in use) raise machine risk. **Expected:** Each single failure flips the Mac **Not compliant** on its own — confirming all four signals are wired in, not just one.

Test 6 — Policy scoping is macOS-only

Do: From a Windows or mobile device (same pilot user), sign in to the same resource. **Expected:** `CA - macOS - Require Compliant Device` does **not** apply (Device platforms = macOS) — proving this policy touches only Macs.

Test 7 — Break-glass / exclusions are exempt

Do: Sign in with a break-glass account (or an account in `macOS-ZeroTrust-Exclusions`). **Expected:** The macOS compliant-device policy does **not** apply — the account is excluded and is never locked out.

Test 8 — Grace period behaves as configured

Do: With a grace period set, make a Compliant Mac non-compliant and watch the status over the grace window. **Expected:** The Mac shows **In grace period** (still allowed) until the window expires, then flips to **Not compliant** and is blocked — confirming users get a warning window, not an instant lockout.

Test 9 — Access-decision report

Do: Run `MACCA-Report.ps1` across the pilot. **Expected:** A clear per-Mac list of compliance state and allow/deny decision, with counts — the artifact you use to sign off each rollout wave.

Pass/fail gate

Gate	Requirement
Pilot pass (Report-only)	Tests 1–7 behave as expected; compliant Macs would be allowed, non-compliant would be blocked, **no unexpected** would-blocks for healthy Macs

Gate	Requirement
Enforcement pass	After enabling On: compliant Macs work, a non-compliant Mac is actually blocked, break-glass verified, grace period behaves (Test 8), no unintended lockouts

6. Troubleshooting Guide — macOS Compliance & Conditional Access for Zero-Trust Access

Use `MACCA-Report.ps1` to see, per Mac, the compliance state and access decision; `MACCA-Validation.ps1` before any enforcement change; and Entra → **Sign-in logs** → **(a sign-in)** → **Report-only / Conditional Access** tabs to see exactly which policy applied and why.

A. Lockout risk / users blocked unexpectedly

Symptom	Cause	Remediation
Legitimate Mac user blocked	The Mac is **Not compliant** (a signal failed)	Run `MACCA-Report.ps1`; fix the failing signal (below); confirm Intune shows **Compliant**
Admin/service account blocked	Break-glass / exception not excluded from the CA policy	Sign in with a **break-glass** account; add the exclusion to `CA - macOS - Require Compliant Device`; re-verify with `MACCA-Validation.ps1`
Whole pilot blocked after enabling On	Policy enabled before Report-only was clean	Use break-glass to set `CA - macOS - Require Compliant Device` back to **Report-only**; resolve would-blocks, then re-enable

B. A specific compliance signal is failing

Signal	Typical cause	Remediation
Encryption / FileVault	FileVault off, or key escrow not complete	Ensure the Intune **Disk encryption (FileVault)** policy targets the Mac; have the user enable FileVault; wait for re-evaluation
Minimum OS version	Mac below the floor you set	Lower the floor to what the fleet can reach *or* update the Mac (Software Update / update policy); confirm with `MACCA-Discovery.ps1`
Firewall	macOS firewall disabled	Enable the firewall (or deploy a settings policy that does); re-sync

Signal	Typical cause	Remediation
Machine risk (Defender)	Active/high-severity detection, or connector off	Resolve the Defender alert on the device; confirm the **Intune** ↔ **Defender** connector is **Enabled**; if Defender isn't deployed, remove the machine-risk rule until it is

C. Compliance state not updating

• The Mac hasn't synced → Company Portal → **check status**, or wait for the next sync cycle; then re-run `MACCA-Report.ps1`.

• Policy not assigned → confirm `macOS - Zero-Trust Compliance` targets a group the Mac is in (`MACCA-Set-ScopeTargeting.ps1 -WhatIf` previews membership).

D. CA policy isn't applying to Macs

• **Device platforms** not set to macOS, or set to something else → confirm **Conditions → Device platforms → macOS** on `CA - macOS - Require Compliant Device`.

• Wrong target resource → the user's app isn't in the policy's **Target resources** (you scoped to Office 365) → widen resources or test against Office 365.

• Policy still in **Report-only** → it only logs; move **Enable policy** → **On** to enforce.

E. "Require compliant device" blocks a device that looks compliant

• Intune shows **Compliant** but Entra still blocks → the Mac may not be **Entra-registered**, so Entra can't read its state → run `MACCA-Validation.ps1`; re-register/re-enroll the Mac.

• Compliance changed very recently → allow time to propagate to Entra; re-test.

F. Report-only shows nothing

• No pilot sign-ins yet for the targeted resource → have pilot users actually sign in from their Macs; wait a few minutes for logs to populate.

G. Diagnostics quick reference

• **Intune** → **Devices** → **Compliance** → `macOS - Zero-Trust Compliance` → **Device status** — per-Mac, per-signal pass/fail.

• **Entra** → **Conditional Access** → **What If** — simulate a macOS sign-in to predict the outcome before enforcing.

- `MACCA-Report.ps1` — the single artifact tying compliance state to the access decision.

H. When to escalate

If a Mac is Intune-**Compliant** and Entra-registered but sign-in is still blocked with the Report-only / Conditional Access tabs inconsistent, capture the sign-in **correlation ID** and open a Microsoft support case.

7. Best Practices — macOS Compliance & Conditional Access for Zero-Trust Access

Grace & rollout safety

- **Use a compliance grace period.** Set **Actions for noncompliance** → **Mark device noncompliant** to fire after a **grace window** (e.g. a few days) rather than immediately, so a Mac that briefly fails a signal gets a warning before access is cut. Start tighter in pilot, more generous during fleet expansion, then tighten once stable.
- **Send the noncompliance email.** Add **Send email to end user** so people know *why* they're about to be blocked and how to fix it — this dramatically reduces help-desk load at cutover.
- **Always pilot the CA policy in Report-only.** Never enable `CA - macOS - Require Compliant Device` straight to On — review the would-be impact in sign-in logs first.
- **Prove the block before you enforce.** Deliberately make one test Mac non-compliant and confirm the Report-only "would block" *before* flipping enforcement On.

Zero-Trust rollout

- **Start narrow, expand in waves.** Pilot group → wave 1 → wave 2 → all Macs. Validate each wave with `MACCA-Report.ps1` before widening.
- **Start with Office 365, then broaden resources.** Scope the CA policy to **Office 365** first; expand to **All cloud apps** only after the compliant-device gate is proven stable.
- **Keep the policy macOS-scoped.** Leave **Conditions** → **Device platforms = macOS** so this policy never surprises Windows or mobile sign-ins — pair it with the general Conditional Access solution for those.
- **Layer signals, don't rush them.** If Defender on macOS isn't ready, ship the encryption + OS + firewall gate now and add the **machine-risk** rule when the connector is on. Tighten the risk threshold (Medium → Low) only after the fleet is clean.

- **Set a reachable OS floor.** Base **Minimum OS version** on what `MACCA-Discovery.ps1` shows your

fleet can actually run; raise it as you patch, not before.

Safety & hygiene

- **Break-glass first, excluded always.** Keep emergency-access accounts (from the Conditional Access

Integration solution) excluded from this policy; re-verify the exclusion before every enforcement change with `MACCA-Validation.ps1`.

- **Name objects clearly.** `macOS - Zero-Trust Compliance`, `CA - macOS - Require Compliant Device`,

`macOS-ZeroTrust-Pilot` — consistent names make audits and handovers trivial.

- **Ensure a remediation path exists.** Before requiring a signal, make sure users *can* satisfy it —

a FileVault disk-encryption policy, an OS-update policy, firewall configuration — so "non-compliant" is always fixable, never a dead end.

- **Monitor continuously.** Review sign-in logs and `MACCA-Report.ps1` after every change; watch for

unexpected denies for legitimate Macs.

- **Use `-WhatIf` first.** Run `MACCA-Remediation.ps1` and `MACCA-Set-ScopeTargeting.ps1` in their

default `-WhatIf` mode and read the preview before committing any change.

8. Deliverables Summary — macOS Compliance & Conditional Access for Zero-Trust Access

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	macOS compliance signals → Conditional Access; the Zero-Trust model for Macs
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps — build compliance, map to a Report-only CA policy, validate, expand
5. Testing Guide	Validation procedures with pass/fail gates (compliant → allowed; non-compliant → denied)
6. Troubleshooting Guide	Field issues, failing signals, and lockout recovery

Document	Purpose
7. Best Practices	Grace periods and safe Zero-Trust rollout

PowerShell scripts

Script	What it does
`MACCA-Discovery.ps1`	Inventories enrolled Macs and their macOS compliance policies + CA policies (Microsoft Graph, read-only)
`MACCA-Validation.ps1`	Confirms pilot Macs are Compliant + Entra-registered and break-glass accounts are excluded — the pre-enforcement check
`MACCA-Report.ps1`	Per-Mac compliance state mapped to the would-be / actual access decision (allow/deny)
`MACCA-Remediation.ps1`	Safe, `-WhatIf`-by-default fixes for common compliance gaps (e.g. re-sync, re-target a Mac)
`MACCA-Set-ScopeTargeting.ps1`	Builds/updates the pilot & rollout group membership from the Mac inventory (`-WhatIf` by default)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the standard model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Named objects this solution creates

- Compliance policy: `macOS - Zero-Trust Compliance`
- Conditional Access policy: `CA - macOS - Require Compliant Device`
- Pilot group: `macOS-ZeroTrust-Pilot` (optional exclusions group `macOS-ZeroTrust-Exclusions`)

Outcome

Only compliant, trusted Macs reach corporate resources — Mac-specific compliance signals (encryption, OS version, firewall, Defender risk) wired to a named Conditional Access policy under Zero-Trust — proven on a pilot in Report-only before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

MACCA-Discovery.ps1

Purpose: Read-only baseline for the macOS Zero-Trust gate. Lists Intune-enrolled Macs with their compliance state and OS version, the macOS compliance policies in the tenant, and the Conditional Access policies that require a compliant device. Use it to size the pilot, pick a reachable minimum-OS floor, and see what already exists before you build. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACCA-Discovery.ps1 -CsvPath C:\Temp\macca-macs.csv Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All, Policy.Read.All. EXPECTED RESULTS A table of Macs (name, OS version, compliance state), a list of macOS compliance policies, and a list of CA policies using the compliant-device control. Optional CSV of the Macs. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\MACCA-Discovery.ps1 -CsvPath C:\Temp\macca-macs.csv Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All, Policy.Read.All. EXPECTED RESULTS A table of Macs (name, OS version, compliance state), a list of macOS compliance policies, and a list of CA policies using the compliant-device control. Optional CSV of the Macs. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath)

MACCA-Remediation.ps1

Purpose: Nudges non-compliant Macs back toward the gate without touching device configuration destructively. For each non-compliant Mac it can request a fresh Intune sync (so a Mac that has since been fixed re-evaluates) and, optionally, add the Mac's primary user to a named remediation group you use to deliver a fix (FileVault / OS-update policy). Runs in preview only unless you pass -Apply; every action honors -WhatIf. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\MACCA-Remediation.ps1 Apply (request re-sync for non-compliant Macs): powershell -ExecutionPolicy Bypass -File .\MACCA-Remediation.ps1 -Apply Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.ReadWrite.All (Group.ReadWrite.All only if using -RemediationGroup). EXPECTED RESULTS A list of non-compliant Macs and the action that would run (WhatIf) or did run (-Apply): "sync requested". Nothing is changed unless -Apply is supplied. .NOTES White-label — the repeatable way. Read-only by default (preview); mutating actions gated behind -Apply and honor -WhatIf. #> [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([switch]$Apply, [string]$RemediationGroup) # Default to preview: without -Apply, force WhatIf so nothing mutates. if (-not $Apply) { $WhatIfPreference = $true }

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\MACCA-Remediation.ps1 Apply (request re-sync for non-compliant Macs): powershell -ExecutionPolicy Bypass -File .\MACCA-Remediation.ps1 -Apply Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.ReadWrite.All (Group.ReadWrite.All only if using -RemediationGroup). EXPECTED RESULTS A list of non-compliant Macs and the action that would run (WhatIf) or did run (-Apply): "sync requested". Nothing is changed unless -Apply is supplied. .NOTES White-label — the repeatable way. Read-only by default (preview); mutating actions gated behind -Apply

and honor -WhatIf. #> [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([switch]$Apply, [string]$RemediationGroup) # Default to preview: without -Apply, force WhatIf so nothing mutates. if (-not $Apply) { $WhatIfPreference = $true }

MACCA-Report.ps1

Purpose: The sign-off artifact for each rollout wave. For every enrolled Mac it reports the Intune compliance state and the resulting access decision under the compliant-device Conditional Access policy: Compliant + registered => ALLOW, otherwise => DENY. Also summarizes recent macOS sign-ins and their Conditional Access result so you can see the gate working live. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACCA-Report.ps1 -CsvPath C:\Temp\macca-decisions.csv Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.Read.All, AuditLog.Read.All, Directory.Read.All. EXPECTED RESULTS A per-Mac table of compliance + ALLOW/DENY with counts, and a summary of recent macOS sign-in CA outcomes (success / blocked). Optional CSV of the per-Mac decisions. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath)

How to run: powershell -ExecutionPolicy Bypass -File .\MACCA-Report.ps1 -CsvPath C:\Temp\macca-decisions.csv Requires Microsoft Graph SDK and scopes: DeviceManagementManagedDevices.Read.All, AuditLog.Read.All, Directory.Read.All. EXPECTED RESULTS A per-Mac table of compliance + ALLOW/DENY with counts, and a summary of recent macOS sign-in CA outcomes (success / blocked). Optional CSV of the per-Mac decisions. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([string]$CsvPath)

MACCA-Set-ScopeTargeting.ps1

Purpose: Keeps the scope groups behind the compliance and Conditional Access policies accurate. Given a target Entra group (e.g. macOS-ZeroTrust-Pilot), it works out which enrolled Macs should be members and previews the additions before making them. Use it to stand up the pilot, then to widen scope wave by wave during expansion. Runs in preview only unless you pass -Apply; every change honors -WhatIf. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\MACCA-Set-ScopeTargeting.ps1 -GroupName macOS-ZeroTrust-Pilot Preview a filtered wave, then apply: powershell -ExecutionPolicy Bypass -File .\MACCA-Set-ScopeTargeting.ps1 -GroupName macOS-ZeroTrust-Pilot -DeviceNameLike 'LAB-' -Apply Requires Microsoft Graph SDK and scopes: Group.ReadWrite.All, DeviceManagementManagedDevices.Read.All, Device.Read.All. EXPECTED RESULTS A list of Macs that would be added to the group (WhatIf) or were added (-Apply), skipping those already members. Nothing is changed unless -Apply is supplied. .NOTES White-label — the repeatable way. Read-only by default (preview); membership changes gated behind -Apply and honor -WhatIf. #> [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'Medium')] param([Parameter(Mandatory)][string]$GroupName, [string]$DeviceNameLike, [switch]$Apply) # Default to preview: without -Apply, force WhatIf so no membership changes. if (-not $Apply) { $WhatIfPreference = $true }

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\MACCA-Set-ScopeTargeting.ps1 -GroupName macOS-ZeroTrust-Pilot Preview a filtered wave, then apply: powershell -ExecutionPolicy Bypass -File .\MACCA-Set-ScopeTargeting.ps1 -GroupName macOS-ZeroTrust-Pilot -DeviceNameLike 'LAB-' -Apply Requires Microsoft Graph SDK and scopes: Group.ReadWrite.All, DeviceManagementManagedDevices.Read.All, Device.Read.All. EXPECTED RESULTS A list of Macs that would be added to the group (WhatIf) or were added (-Apply), skipping those

already members. Nothing is changed unless -Apply is supplied. .NOTES White-label — the repeatable way. Read-only by default (preview); membership changes gated behind -Apply and honor -WhatIf. #> [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'Medium')] param([Parameter(Mandatory)][string]$GroupName, [string]$DeviceNameLike, [switch]$Apply) # Default to preview: without -Apply, force WhatIf so no membership changes. if (-not $Apply) { $WhatIfPreference = $true }

MACCA-Validation.ps1

Purpose: The check to run before you flip the Conditional Access policy from Report-only to On. Confirms every Mac in the pilot group is Intune-Compliant and Entra-registered (the two facts the compliant-device control depends on), and confirms the break-glass accounts you name are excluded from the CA policy so nobody can be locked out. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\MACCA-Validation.ps1 -PilotGroup macOS-ZeroTrust-Pilot -CaPolicyName 'CA - macOS - Require Compliant Device' -BreakGlassUpns breakglass1@your-tenant.onmicrosoft.com,breakglass2@your-tenant.onmicrosoft.com Requires Microsoft Graph SDK and scopes: Group.Read.All, DeviceManagementManagedDevices.Read.All, Policy.Read.All, User.Read.All. EXPECTED RESULTS Per pilot Mac: COMPLIANT/REGISTERED (good) or the failing fact. Per break-glass account: EXCLUDED (good) or NOT EXCLUDED (fix before enforcing). "READY TO ENFORCE" only when all pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$PilotGroup, [Parameter(Mandatory)][string]$CaPolicyName, [string[]]$BreakGlassUpns)

How to run: powershell -ExecutionPolicy Bypass -File .\MACCA-Validation.ps1 -PilotGroup macOS-ZeroTrust-Pilot -CaPolicyName 'CA - macOS - Require Compliant Device' -BreakGlassUpns breakglass1@your-tenant.onmicrosoft.com,breakglass2@your-tenant.onmicrosoft.com Requires Microsoft Graph SDK and scopes: Group.Read.All, DeviceManagementManagedDevices.Read.All, Policy.Read.All, User.Read.All. EXPECTED RESULTS Per pilot Mac: COMPLIANT/REGISTERED (good) or the failing fact. Per break-glass account: EXCLUDED (good) or NOT EXCLUDED (fix before enforcing). "READY TO ENFORCE" only when all pass. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string]$PilotGroup, [Parameter(Mandatory)][string]$CaPolicyName, [string[]]$BreakGlassUpns)